

WIDIANA ANGGREANI RAHWAL

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SUMMARY

Software developer with experience building web and mobile applications across multiple industries, including oil & gas, healthcare, education, tourism, and retail. Proficient in ASP.NET MVC, Django, Flutter, and Laravel for developing industrial monitoring systems, operational dashboards, automation systems, and transaction applications. Skilled in user requirements analysis, database performance optimization, and building scalable, maintainable, high-performance applications.

WORK EXPERIENCE

Software Developer

January 2024 – Present

PT. Sila Teknologi Utama

Key Achievements:

- Developed 19 functional requirements (9 dashboards & toolboxes, 10 additional features) for a well monitoring system at Pertamina Hulu Mahakam using ASP.NET MVC and MSSQL / Oracle SQL.
- Revamped application performance by migrating business logic and calculations from the application layer to Stored Procedures and SQL Views with parameters and pagination, reducing data transferred per request from millions of records to 50 records — significantly improving access speed.
- Contributed to the development of 10+ applications across multiple sectors (education, healthcare, retail, and tourism) using Django and Flutter with PostgreSQL and Oracle SQL.
- Automated well monitoring systems replacing manual Excel-based processes, improving data analysis speed.
- Developed an integrated ticketing platform and payment system.
- Optimized backend architecture and system security to improve code scalability and maintainability.

Project Details:

- A. Industrial Automation & Monitoring System Development (Client: Pertamina Hulu Mahakam)
- Designed and developed interactive custom charts using Chart.js for visualizing gas, oil, and sand risk assessment data, enabling faster issue identification.
 - Designed an Entity-Relationship Diagram (ERD) to consolidate multiple databases into a centralized data warehouse, improving data structure and control.
 - Developed 19 functional requirements using ASP.NET MVC and MSSQL / Oracle SQL, covering 9 dashboards/toolboxes and 10 supporting features such as data tables, Excel upload, and landing pages.
 - Revamped application performance by moving calculation logic from the application layer to Stored Procedures and SQL Views, reducing data volume per request from millions of records to 50 records through database-level parameter filtering and pagination.
- B. Web Feature Development (Django)
- Optimized the authentication system by implementing OOP principles, Role-Based Access Control (RBAC), and automated activation token delivery via email, improving user registration security.
 - Developed a real-time chat feature using Django Channels, ASGI, and Redis to support stable and responsive two-way communication.
 - Developed a complete payment module (OTS, QRIS, EDC, etc.) with Midtrans integration and a voucher and promo system, contributing to increased transactions.
 - Developed a scheduler using django-crontab to automate daily data synchronization and notification email delivery (alerts, transaction confirmations).
 - Redesigned several pages and developed additional key features.
 - Developed a patient registration module by integrating a queue system, barcode, and online form, reducing registration time and minimizing input errors.
- C. Mobile App Development & Integration (Flutter)
- Designed UI/UX for mobile applications.
 - Developed Flutter applications and REST APIs to ensure reliable real-time data synchronization for transactions, inventory, pricing, tickets, and other data.
 - Developed medical sample identification through a barcode-based labeling system that can be scanned to instantly display patient data.
- D. E-Learning & Assessment System Development
- Designed and built a secure, multi-user Website Exam System with student access restricted through Safe Exam Browser to prevent cheating.
 - Designed and built an automated grading module for various question types, including a weighting and difficulty level system automatically calculated based on student answer percentages.
- E. Training & Documentation
- Conducted system usage training for end users.
 - Produced clear and structured End User Documentation (EUD) as a feature guidebook.

- Conducted User Acceptance Testing (UAT) and System Integration Testing (SIT) on a Laravel-based website and compiled test result documentation.
- Performed bug analysis, communicated findings to the development team, and produced comprehensive testing reports using Trello.

EDUCATION & TRAINING

- Developed an Internal Quality Assurance Information System end-to-end using Personal Extreme Programming methodology, built with Laravel and MySQL to streamline academic accreditation data management and reporting.
- Built various learning projects to master full-stack development using Laravel (backend), React.js (frontend), and MySQL (database).
- Technical: PHP, Laravel, MySQL, React.js, VirtualBox, Docker (Basic).
- Personal Development: Participated in leadership and organizational management training (LKMM Pra-TD, LKMW) and scientific reasoning training (PKTI) to strengthen soft skills.

- Solidified foundation in Object-Oriented Programming (OOP) and clean code practices through best practices implementation and effective debugging techniques.
- Skilled in Flutter application development, covering responsive UI/UX design, state management with Provider, and reliable REST API integration.

- Developed DentsHealth, a dental health diagnostic application built with Kotlin, leveraging Natural Language Processing (NLP) to analyze user complaints and cloud computing for Machine Learning model deployment, resulting in automated detection of dental and oral health issues.
- Learned Android development hard skills including Kotlin, Android application fundamentals, UX design, SOLID principles, and OOP paradigms.
- Developed workplace soft skills including time management, adaptability, critical thinking, and English proficiency in both spoken and written forms.

ORGANIZATIONAL EXPERIENCE

- Designed logos, content, and Instagram feed for HMIF using Adobe Illustrator and CorelDraw.
- Participated as committee member in the HMIF Student Chairman Election.
- Participated as committee member and prepared a CorelDraw design training event for Informatics students.

SKILLS

- **Programming language:** Python, Dart, JavaScript (jQuery), PHP, C#
- **Framework:** Django, Flutter, Laravel, ASP.NET MVC
- **Database:** PostgreSQL, MySQL, Oracle SQL
- **Development Tools:** Git, Postman, Docker (Basic), VirtualBox, XAMPP, Laragon
- **Design & Collaboration Tools:** Figma, Draw.io, Miro, Trello
- **Front-End Tools:** React.js (Basic), Bootstrap, HTML, CSS, Chart.js
- **Non-Technical:** Analytical thinking, time management, teamwork & collaboration, adaptability, problem solving.
- **Language:** Indonesian (Native), English (Intermediate | IAET ITK: 520, TOEFL Prediction: 610)

ACHIEVEMENTS & OTHER EXPERIENCES

- **Copyright & Certificate:** Software Copyright for PsyCare, 15+ technology certifications from Dicoding, IDCamp, Google Bangkit, and other programs including Flutter, Android Development, Dart, SOLID Principles, and Fundamental AI.
- **Other Projects:** CISA (Covid-19 Information and Self Check Up Application), PetYourself, PsyCare, SIKMA (Internship Information System), DentsHealth, Restaurant App, SIMJAMU (Quality Assurance System), LeafPad, Gallery App.
- **Volunteer Work:** Mangrove Warrior 2025, Lestravida Balikpapan Batch 1.